Dept. of Computer and Communication Engineering

Patuakhali Science and Technology University

5th Semester (Level-3, Semester-I), Mid Examination of B.Sc. Engg. (CSE), January-June: 2022

Course Code: CCE313 Course Title: Computer Networks

Credit Hour: 3.0   Full Marks: 15   Time: 60 minutes

24

01. Find the netid, subnet mask and the hostid of the following IP addresses: **2**
a. 114.34.2.8 A b. 132.56.8.6 B c. 208.34.54.12 C d. 251.34.98.5 E

02. An address in a block is given as 191.8.243.9. Find the number of addresses in the block, subnet mask the first address, and the last address. **3**

03. A company is granted the site address 201.70.64.0. The company needs six subnets. Design the subnets and shows the first address and last address with subnetmask through a pictorial representation. **3**

04. What are the services that a transport-layer protocol cannot provide to applications invoking it? **2**
throughput, security, time, integrity

05. How can an application use the services of TLS? TCP **2**

06. Describe how installing a proxy server can reduce the delay in receiving a requested object. **3**

192- 239
224- 255
240-

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Mid: I   Semester: 5th   Batch : 16th   Marks 15   Session 2018-2019

Course Code: CCE 313   Time: 55 Min   Course Title: Computer Network

1. An classful address in a block is given as 222.8.17.9. Find the number of addresses in the block, the first address, and the last address. Draw also the block diagram of this IP address topology. 16 3
2. Suppose you have given a classful block of IP 140.15.0.0. Now you need to divide this IP block to four subnetwork with equal IP address space of each block. Now extracting the first address last address, subnetwork mask to follow the proper procedure and also draw the diagram of the sub network. 5
3. Suppose Alice, who always accesses the Web using Internet Explorer from her home PC, contacts Amazon.com for the first time. Let us also suppose that in the past she has already visited the eBay site. Now, what will happen, when the request comes into the Amazon Web server for the first time and then one week later? Illustrate the communication process between Alice's browser and the Amazon web server with respect to cookies. 4
4. What is the function of conditional GET? 3

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Mid: I Semester: 5^{th} Batch : 15^{th} Marks 15

Course Code: CCE 313 Time: 50 Min Course Title: Computer Network

1 BTCL is granted a block of addresses starting with 254.103.0.0/15. The BTCL needs to distribute these addresses to three division of customers as follows: 10

i) The Dhaka city has 64 customers; each needs approximately 256 addresses.

ii) The Barisal city has 128 customers; each needs approximately 128 addresses.

iii) The Khulna city has 128 customers; each needs approximately 256 addresses.

Now design the subblocks and find out how many addresses are still available after these allocations with diagram.

2 One of the addresses in a block is 167.199.170.82/24. Find the number of addresses in the network, the first address, and the last address with proper procedure. 5

Course Code: CCE 313 Course Title: Computer Networks Time: 45 minutes

1. What is the network address in a block of addresses? How can we find the network address if one of the addresses in a block is given? 3
2. An ISP is granted a block of addresses starting with 120.60.4.0/22. The ISP wants to distribute these blocks to 100 organizations with each organization receiving just eight addresses. Design the subblocks and give the slash notation for each subblock. Find out how many addresses are still available after these allocations. 5
3. In a connection, the value of *cwnd* is 2500 and the value of *rwnd* is 4500. The host has sent 2000 bytes which has not been acknowledged. How many more bytes can be sent? 2
4. A window holds bytes 2001 to 6000. The next byte to be sent is 3001. Draw a figure to show the situation of the window after the following two events: 4
   a. An ACK segment with the acknowledgement number 3500 and window size advertisement 4000 is received.
   b. A segment carrying 1500 bytes is sent.
5. What is a socket address? 1